

Proctial - 1 - Reinforcement Learning

(58)

81)

A — 3, 2, 4

B — 6, 7

C — 5, 4, 6

$$Q(a) = \frac{\text{Sum of rewards}}{N(a)}$$

$$A \rightarrow N(A) = 3$$

$$3 + 2 + 4 = 9$$

$$Q(A) = \frac{9}{3} = 3$$

$$B \rightarrow N(B) = 2$$

$$6 + 7 = 13$$

$$Q(B) = \frac{13}{2} = 6.5$$

$$C \rightarrow N(C) = 3$$

$$5 + 4 + 6 = 15$$

$$Q(C) = \frac{15}{3} = 5$$

Greedy Selected Machine = B

Q2)

$$\text{Initially : } Q(A) = Q(B) = Q(C) = 0$$

$$N(A) = N(B) = N(C) = 0$$

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{R - Q_{\text{old}}(a)}{N(a)}$$

Machine A
first reward = 3

$$Q(A) = 0 \quad N(A) = 0$$

$$N(A) = 1$$

$$Q(A) = 0 + \frac{3 - 0}{1}$$

$$Q(A) = 3$$

second reward = 2

$$N(A) = 2$$

$$Q(A) = 3 + \frac{2 - 3}{2}$$

$$= 3 - \frac{1}{2}$$

$$Q(A) = 2.5$$

third reward = 4

$$N(A) = 3$$

$$Q(A) = 2.5 + \frac{4 - 2.5}{3}$$

$$= 2.5 + 0.5$$

$$Q(A) = 3$$

$$N(A) = 3, Q(A) = 1$$

Machine B

first reward = 6

$$N(B) = 1$$

$$Q(B) = 0 + \frac{6-0}{1}$$

$$Q(B) = 6$$

second reward = 7

$$N(B) = 2$$

$$Q(B) = 6 + \frac{7-6}{2}$$

$$= 6 + \frac{1}{2}$$

$$= 6.5$$

Machine C

first reward = 5

$$N(C) = 1$$

$$Q(C) = ~~0 + \frac{5-0}{1}~~ 0 + \frac{5+0}{1}$$

$$Q(C) = 5$$

second reward = 4

$$N(C) = 2$$

$$Q(C) = 5 + \frac{4-5}{2}$$

$$= 5 - \frac{1}{2}$$

$$Q(C) = 4.5$$

Therol reward = 5

$$N(c) = 3$$

$$Q(c) = 4.5 + \frac{6 - 4.5}{3}$$

$$= 4.5 + \frac{1.5}{3}$$

$$= 4.5 + 0.5$$

$$Q(c) = 5$$

$$Q(B) = 4.5$$

greedy selected machine = B.